

Center for Organ Recovery & Education (CORE)

Executive Summary

Center for Organ Recovery & Education (CORE)

Bill DiPlacido, Cade Gallagher

Student Consulting Team

Matthew Wong, Alex Torres Vivaldo, Josh Dong

Background

The Center for Organ Recovery & Education (CORE) is a regional organ procurement organization responsible for coordinating organ, tissue, and cornea recovery for transplantation. Operating in a highly regulated and time-sensitive healthcare environment, CORE relies on accurate, real-time data to manage donor cases and ensure effective communication across hospitals, transplant centers, and internal teams. Administrators on Call (AOCs), who oversee donor case activity during 24-hour shifts, depend heavily on TrackerBoard, CORE's central operational system. However, limitations in mobile usability created workflow inefficiencies, increased reliance on workarounds, and introduced risks to data accuracy and continuity.

Project Description

Project Opportunity

CORE identified a critical need to improve mobile access to TrackerBoard for AOCs, who must frequently monitor and update donor case information throughout their shifts. The existing system's reliance on desktop workflows limited flexibility and contributed to inefficiencies, delayed updates, and potential data inconsistencies. The opportunity was to create a mobile-first solution that enables AOCs to securely access and update critical information in real time without being tied to a laptop.

Project Vision

The project aimed to deliver a mobile-first, role-aware minimum viable product (MVP) that supports CORE's most critical AOC workflows within TrackerBoard. Rather than rebuilding the system, the solution extends CORE's existing Microsoft-based architecture to provide secure, real-time mobile functionality. The vision emphasized usability, reliability, and alignment with existing workflows, while creating a scalable foundation for future enhancements.

Project Outcomes

- Technical Outcomes: Established a mobile integration foundation aligned with CORE's existing systems, including API development, Entra-based authentication, and backend architecture for secure mobile access.
- Workflow & UX Outcomes: Validated and refined mobile workflows for AOCs, including Active Organ Cases, Report Out, and essential form updates. Introduced mobile-friendly interaction patterns to replace desktop-specific behaviors.

- Capacity & Handoff Outcomes: Strengthened collaboration with CORE stakeholders and defined a clear transition plan, including documentation and a designated handoff lead.
- Top-Level Outcomes: Clarified MVP scope, reduced ambiguity in mobile workflows, and positioned CORE to move from concept to pilot implementation with a realistic and maintainable solution.

Project Deliverables

The final deliverables include a mobile TrackerBoard MVP focused on high-priority AOC workflows, supported by backend integration and authentication infrastructure. Additional deliverables include workflow designs, prototypes, validation artifacts, and a comprehensive handoff package with documentation covering system setup, architecture decisions, and usage guidelines. Together, these deliverables provide CORE with both a functional starting point and the resources needed for continued development and deployment.

Recommendations

CORE should treat this MVP as the foundation for a phased mobile rollout. Immediate next steps include conducting real-world testing with AOCs, refining workflows based on live usage, and ensuring system reliability. CORE should also prioritize clear documentation, maintain strong role-based access controls, and address underlying system stability issues such as data consistency and connectivity. Future development should proceed incrementally, expanding functionality only after the MVP proves effective in practice.

Student Consulting Team

Matthew Wong led the project's backend architecture, mobile API integration, and authentication work. He defined the mobile API boundary, implemented Entra / bearer-auth integration and server-side role authorization, and connected the mobile app to real backend endpoints instead of mock data. Matthew is a junior majoring in Information Systems, currently working as a Software Engineer at Automattic (on the Beeper team) and is looking toward a career building high-impact software systems at startups.

Alexander Torres Vivaldo served as project manager and quality-assurance lead while also contributing heavily to the frontend mobile MVP. He organized agendas and meeting documentation, reviewed technical work, and helped build and document the mobile prototype, including setup materials for CORE's future use. Alex is a junior majoring in Information Systems, and is an aspiring software engineer.

Josh Dong led stakeholder research and workflow refinement for the mobile experience. He conducted AOC interviews, created research materials, and translated user feedback into improvements to the AOC form structure, filtering behavior, and overall mobile workflow design. Josh is a junior majoring in Information Systems, currently a business owner and is an aspiring technical product manager/founder.